

Research Questions and Manuscript Navigation

Research Questions and Manuscript Navigation I

This paper addresses eleven core research questions, each developed in specific sections and yielding a principal contribution (**C1–C11**, as enumerated in ??). The table below provides a navigational overview; the reading guide that follows explains the logical dependencies between questions.

Table 1: Core research questions mapped to corresponding manuscript sections and resulting contributions.

#	Research Question	Addressing Sec- tions	C
RQ1	Can linguistic case systems—across all major typological alignment patterns—be formalized within a single algebraic framework, with roles as objects and grammatical relations as morphisms?	??, ??	C1
RQ2	Do string diagrams derived from pregroup type reductions provide a computationally and cognitively superior representation of case derivations compared to linear notation?	??, ??, ??	C2

Research Questions and Manuscript Navigation II

RQ3	Can the DisCoCat meaning functor be extended with case-typed noun spaces to unify formal and distributional semantics—and does this unification generalize to discourse via DisCoCirc?	??, ??, ??	C3
RQ4	Does equipping case categories with $[0, 1]$ -enriched hom-values yield a principled bridge between symbolic grammar and statistical semantics, and can categorical magnitude serve as a complexity invariant?	??, ??	C4
RQ5	Can classifying toposes and Morita equivalence (or explicit bridge toposes) scaffold transfer of topos-expressible invariants between distinct formalizations of case theory (typological, type-logical, distributional, enriched), when equivalence is exhibited?	??	C5
RQ6	Are commutative diagrams <i>cognitively privileged</i> representations—recruiting spatial reasoning, free-ride inference, and predictive processing—beyond being merely convenient notation?	??, ??	C6

Research Questions and Manuscript Navigation III

RQ7	Can the entire categorical framework be computationally verified with real (non-mock) automated tests at 90% coverage, confirming that the abstractions are executable?	??, ??	C7
RQ8	Do categorical string diagrams extend naturally into quantum generalizations via TQNNs, ZX-calculus, and sheaf-theoretic quantum semantics—and does quantum entanglement provide genuine advantages for semantic communication?	??, ??	C8
RQ9	Can prompt injection attacks be formulated mathematically as illicit cross-category morphisms (type violations), yielding a decidable static check relative to a fixed protocol / interaction category (compile-time where that grammar is enforced)?	??	C9
RQ10	Does integrating enriched category theory with active inference generate <i>falsifiable</i> neurolinguistic predictions (P600/N400 amplitudes scaling with enriched morphism weights)?	??, ??	C10

Research Questions and Manuscript Navigation IV

RQ11	Can grammatical case roles (NOM, ACC, INS) rigorously type-check the tensor payloads of multi-agent AI framework protocols (A2A, MCP, ANP) to prevent agentic hijacking?	??	C11
-------------	--	-----------	------------

Research Questions and Manuscript Navigation V

Reading guide. **RQ1–RQ5** build the mathematical framework cumulatively: each layer depends on the preceding one, from case categories (RQ1) through string diagrams (RQ2), distributional semantics (RQ3), enriched structure (RQ4), to topos-theoretic transfer (RQ5). **RQ6** provides the cognitive science meta-argument that threads through the entire paper, grounding diagrams in spatial reasoning and predictive processing. **RQ7** supplies computational validation of the framework via automated testing. **RQ8–RQ11** extend the framework into four application domains: quantum semantics (RQ8), cognitive security (RQ9), falsifiable neurolinguistic predictions (RQ10), and multi-agent AI protocols (RQ11). The conclusion (??) revisits each question with a summary of results and identifies eight open directions (**F1–F8**).